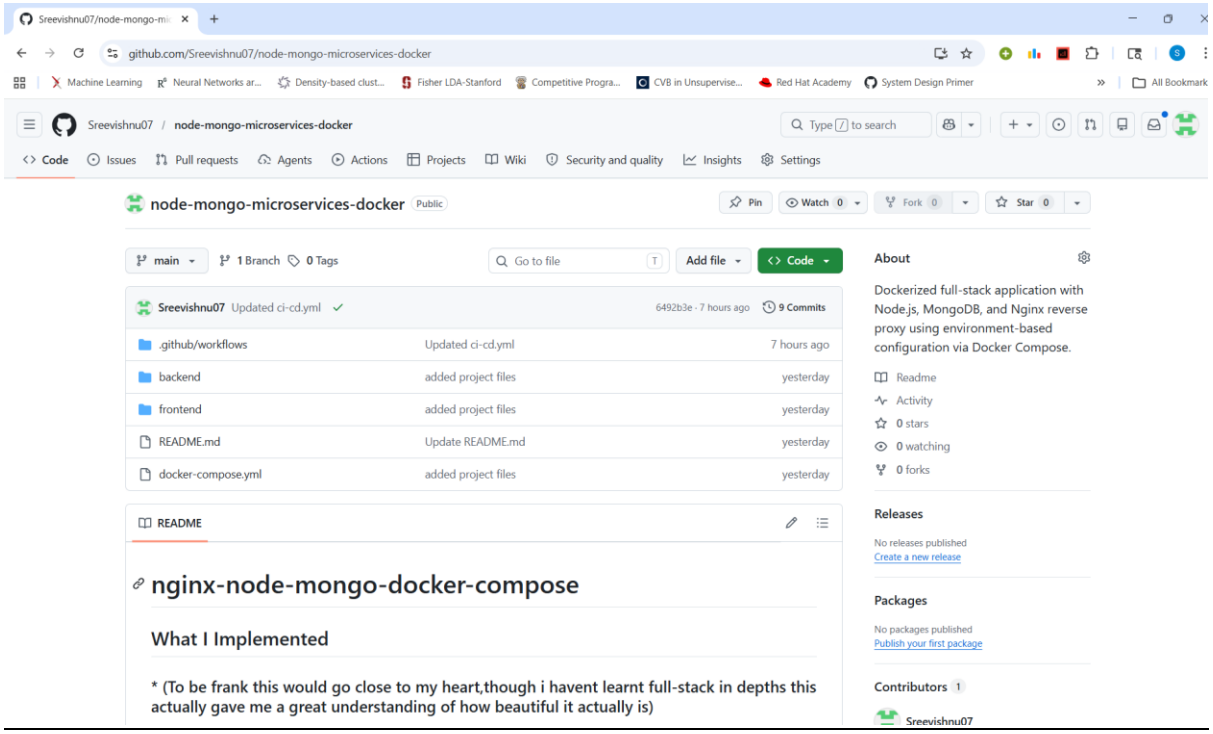
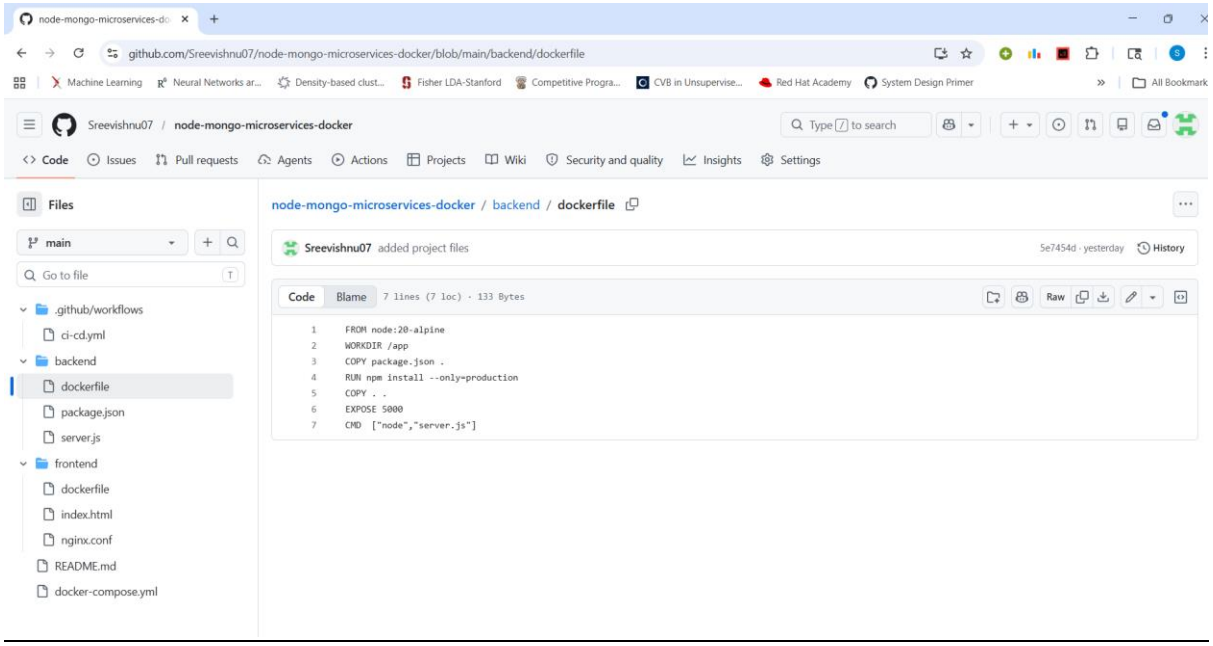
# Production-Ready Dockerized Microservices with Nginx, Node.js, MongoDB & CI/CD Pipeline



node-mongo-microservices-do

github.com/Sreevishnu07/node-mongo-microservices-docker/blob/main/backend/package.json

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Code | Blame | 9 lines (9 loc) · 143 Bytes | Raw

```
1  {
2    "name": "backend",
3    "version": "1.0.0",
4    "main": "server.js",
5    "dependencies": {
6      "express": "^4.18.2",
7      "mongodb": "^6.0.0"
8    }
9  }
```

---

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github.com/Sreevishnu07/node-mongo-microservices-docker/blob/main/backend/server.js

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Code | Blame | 50 lines (42 loc) · 1018 Bytes | Raw

```
1  const express = require("express");
2  const { MongoClient } = require("mongodb");
3
4  const app = express();
5
6  const {
7    DB_HOST,
8    DB_PORT,
9    DB_USER,
10   DB_PASSWORD,
11   DB_NAME,
12   PORT
13 } = process.env;
14
15 const uri = `mongodb://${DB_USER}:${DB_PASSWORD}@${DB_HOST}:${DB_PORT}/${DB_NAME}?authSource=admin`;
16
17 let db;
18 async function connectDB() {
19   while (true) {
20     try {
21       const client = new MongoClient(uri);
22       await client.connect();
23       db = client.db(DB_NAME);
24       console.log("Connected to MongoDB");
25       break;
26     } catch (err) {
27       console.log("DB not ready, retrying...");
28       await new Promise(res => setTimeout(res, 3000));
```

---

- package.json
- server.js
- frontend
  - dockerfile
  - index.html
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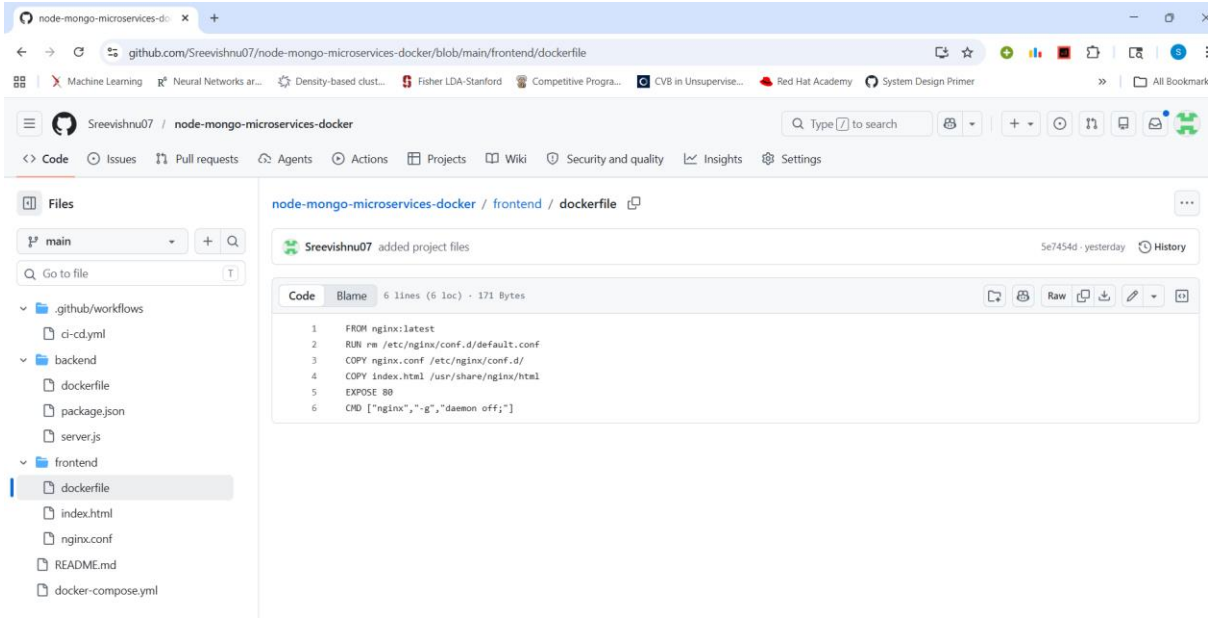
```
29     }
30   }
31 }
32
33 connectDB();
34
35 app.get("/", async (req, res) => {
36   try {
37     const collections = await db.listCollections().toArray();
38
39     res.json({
40       message: "Backend is running",
41       collections
42     });
43   } catch (err) {
44     res.status(500).json({ error: "DB not ready yet" });
45   }
46 });
47
48 app.listen(PORT, () => {
49   console.log(`Server running on port ${PORT}`);
50 });
```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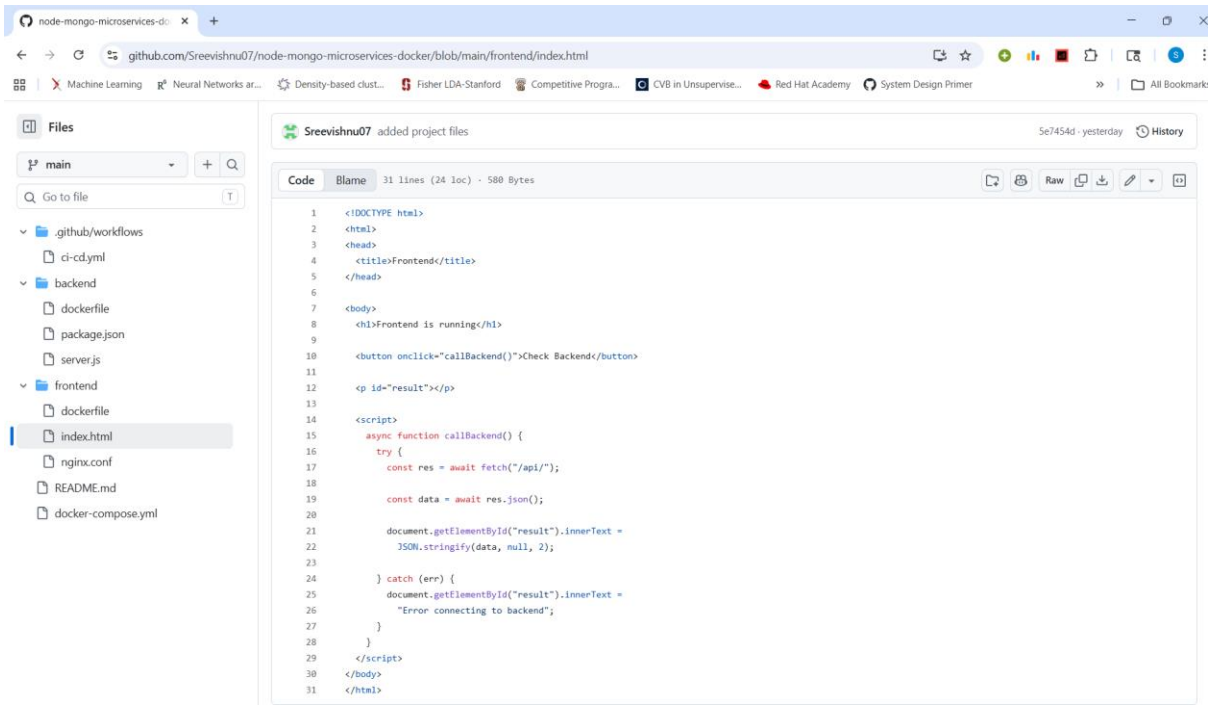
6 lines (6 loc) · 171 Bytes
Raw
FROM nginx:latest
RUN rm /etc/nginx/conf.d/default.conf
COPY nginx.conf /etc/nginx/conf.d/
COPY index.html /usr/share/nginx/html
EXPOSE 80
CMD ["nginx","-g","daemon off;"]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31 lines (24 loc) · 580 Bytes
Raw
<!DOCTYPE html>
<html>
<head>
<title>Frontend</title>
</head>
<body>
<h1>Frontend is running</h1>
<button onclick="callBackend()">Check Backend</button>

<script>
async function callBackend() {
try {
const res = await fetch("/api/");
const data = await res.json();
document.getElementById("result").innerText =
JSON.stringify(data, null, 2);
} catch (err) {
document.getElementById("result").innerText =
"Error connecting to backend";
}
}
</script>
</body>
</html>

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github.com/Sreevishnu07/node-mongo-microservices-docker/blob/main/frontend/nginx.conf

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node-mongo-microservices-docker / frontend / nginx.conf

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Code | Blame — 14 lines (13 loc) · 278 Bytes — Raw

```
server{
  listen 80;
  location / {
    root /usr/share/nginx/html;
    index index.html;
  }
  location /api/ {
    proxy_pass http://backend:5000/;

    proxy_http_version 1.1;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
  }
}
```

---

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github.com/Sreevishnu07/node-mongo-microservices-docker/blob/main/docker-compose.yml

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Code | Blame — 36 lines (33 loc) · 666 Bytes — Raw

```
services:
  mongodb:
    image: mongo:7
    container_name: mongodb
    restart: unless-stopped
    env_file:
      - .env
    volumes:
      - mongo_data:/data/db
    healthcheck:
      test: ["CMD","mongosh","--eval","db.runCommand({ping: 1})"]
      interval: 5s
      timeout: 3s
      retries: 5

  backend:
    build: ./backend
    container_name: backend3
    restart: unless-stopped
    env_file:
      - .env
    ports:
      - "5001:5000"
    depends_on:
      mongodb:
        condition: service_healthy

  frontend:
    build: ./frontend
    container_name: frontend3
    restart: unless-stopped
    ports:
      - "3000:80"

volumes:
  mongo_data:
```

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View Runs

Sreevishnu07 Updated ci-cd.yml ✓ 6492b3e · 7 hours ago History

Code | Blame | 137 lines (114 loc) · 3.69 KB | Raw

```yaml
name: Full CI-CD Pipeline

on:
  push:
    branches:
      - main

jobs:
  test:
    runs-on: ubuntu-latest
    services:
      mongodb:
        image: mongo:7
        ports:
          - "27017:27017"
        env:
          MONGO_INITDB_ROOT_USERNAME: root
          MONGO_INITDB_ROOT_PASSWORD: root
          MONGO_INITDB_DATABASE: my_db
    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Setup Node
        run: |
          cd backend
```

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github.com/Sreevishnu07/node-mongo-microservices-docker/blob/main/.github/workflows/ci-cd.yml

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```yaml
      - name: Wait for Backend
        run: |
          for i in {1..15}; do
            if nc -z 127.0.0.1 27017; then
              echo "MONGO DB ready"
              exit 0
            fi
            echo "Waiting for connection.."
            sleep 2
          done
          echo "Mongo failed"
          exit 1

      - name: Start Backend and test api
        env:
          DB_HOST: 127.0.0.1
          DB_PORT: 27017
          DB_USER: root
          DB_PASSWORD: root
          DB_NAME: my_db
          PORT: 5000
        run: |
          node backend/server.js &
          for i in {1..10}; do
            if curl -f http://127.0.0.1:5000; then
              echo "API working"
              exit 0
            fi
            sleep 2
          done
          echo "Backend failed"
          exit 1

  integration-test:
    runs-on: ubuntu-latest
```

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```yaml

    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Create .env file
        run: |
          echo "MONGO_INITDB_ROOT_USERNAME=admin">>.env
          echo "MONGO_INITDB_ROOT_PASSWORD=${{ secrets.MDB_PASS }}">>.env
          echo "MONGO_INTDB_DATABASE=my_db">>.env
          echo "DB_HOST=mongodb">>.env
          echo "DB_PORT=27017">>.env
          echo "DB_USER=admin">>.env
          echo "DB_PASSWORD=${{ secrets.MDB_PASS }}">>.env
          echo "DB_NAME=my_db">>.env
          echo "PORT=5000">>.env
          echo "NODE_ENV=production">>.env

      - name: Start full stack
        run: docker compose up -d

      - name: Wait for services
        run: sleep 15

      - name: Test api
        run: curl -f http://127.0.0.1:3000/api/ || exit 1

      - name: Shutdown
        run: docker compose down --remove-orphans

  build-push:
    runs-on: ubuntu-latest
    needs: integration-test
```

node-mongo-microservices-do

github.com/Sreevishnu07/node-mongo-microservices-docker/blob/main/.github/workflows/ci-cd.yml

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Raw

```yaml
  build-push:
    runs-on: ubuntu-latest
    needs: integration-test

    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: set image tag
        id: vars
        run: echo "TAG-${{ github.sha }}" >> $GITHUB_OUTPUT

      - name: Docker login
        run: echo "${{ secrets.DOCKER_PASSWORD }}" | docker login -u ${{ secrets.DOCKER_USERNAME }} --password-stdin

      - name: Build backend image
        run: |
         docker build \
         --cache-from ${{ secrets.DOCKER_USERNAME }}/backend:latest \
          -t ${{ secrets.DOCKER_USERNAME }}/backend:latest \
          -t ${{ secrets.DOCKER_USERNAME }}/backend:${{ steps.vars.outputs.TAG }} ./backend

      - name: Build frontend image
        run: |
          docker build \
          --cache-from ${{ secrets.DOCKER_USERNAME }}/frontend:latest \
          -t ${{ secrets.DOCKER_USERNAME }}/frontend:latest \
          -t ${{ secrets.DOCKER_USERNAME }}/frontend:${{ steps.vars.outputs.TAG }} ./frontend

      - name: push frontend image
        run:  |
           docker push ${{ secrets.DOCKER_USERNAME }}/frontend:latest
           docker push ${{ secrets.DOCKER_USERNAME }}/frontend:${{ steps.vars.outputs.TAG }}
```

- frontend
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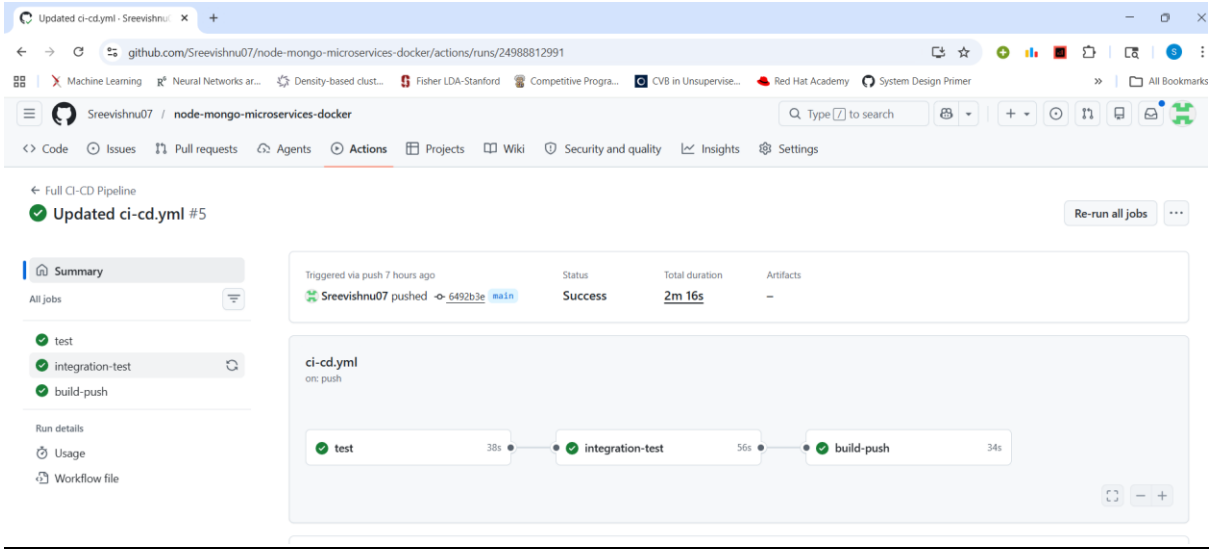
```yaml
          -t ${{ secrets.DOCKER_USERNAME }}/backend:${{ steps.vars.outputs.TAG }} ./backend

      - name: Build frontend image
        run: |
          docker build \
          --cache-from ${{ secrets.DOCKER_USERNAME }}/frontend:latest \
          -t ${{ secrets.DOCKER_USERNAME }}/frontend:latest \
          -t ${{ secrets.DOCKER_USERNAME }}/frontend:${{ steps.vars.outputs.TAG }} ./frontend

      - name: push frontend image
        run:  |
           docker push ${{ secrets.DOCKER_USERNAME }}/frontend:latest
           docker push ${{ secrets.DOCKER_USERNAME }}/frontend:${{ steps.vars.outputs.TAG }}
```

```yaml

      - name: backend image
        run: |
            docker push ${{ secrets.DOCKER_USERNAME }}/backend:latest
            docker push ${{ secrets.DOCKER_USERNAME }}/backend:${{ steps.vars.outputs.TAG }}
```


Updated ci-cd.yml - Sreevishnu
github.com/Sreevishnu07/node-mongo-microservices-docker/actions/runs/24988812991
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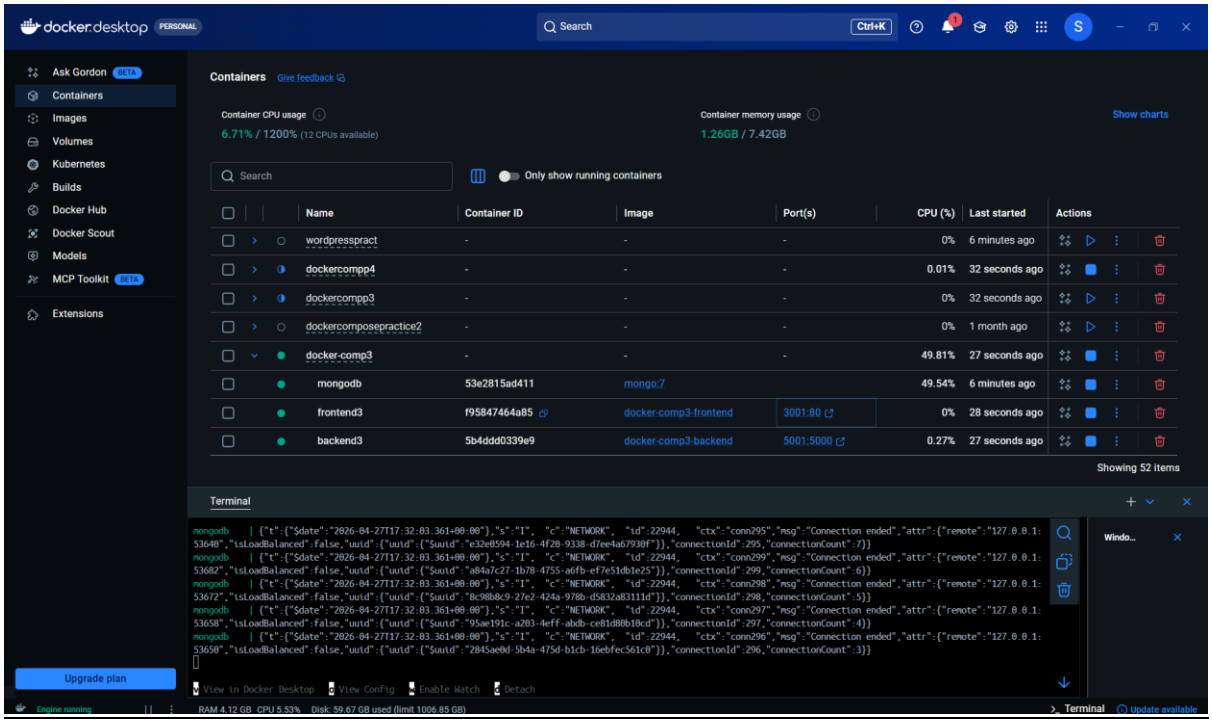
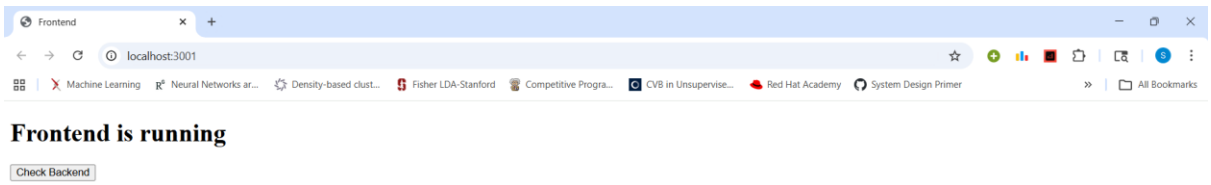
Updated ci-cd.yml #5
Re-run all jobs
Summary
All jobs
test
integration-test
build-push
Run details
Usage
Workflow file
Triggered via push 7 hours ago
Sreevishnu07 pushed 6492b3e main
Status
Success
Total duration
2m 16s
Artifacts
–
ci-cd.yml
on: push
test 38s
integration-test 56s
build-push 34s


docker desktop PERSONAL
Search
Ctrl+K
Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Upgrade plan
Containers
Give feedback
Container CPU usage
6.71% / 1200% (12 CPUs available)
Container memory usage
1.26GB / 7.42GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
wordpresspract
dockercompp4
dockercompp3
dockercomposepractice2
docker-comp3
mongodb
53e2815ad411
mongo:7
frontend3
f95847464a85
docker-comp3-frontend
3001:80
backend3
5b4ddd0339e9
docker-comp3-backend
5001:5000
Showing 52 items
Terminal
View in Docker Desktop
View Config
Enable Watch
Detach
RAM 4.12 GB CPU 5.53% Disk 59.67 GB used (limit 1006.85 GB)
Terminal
Update available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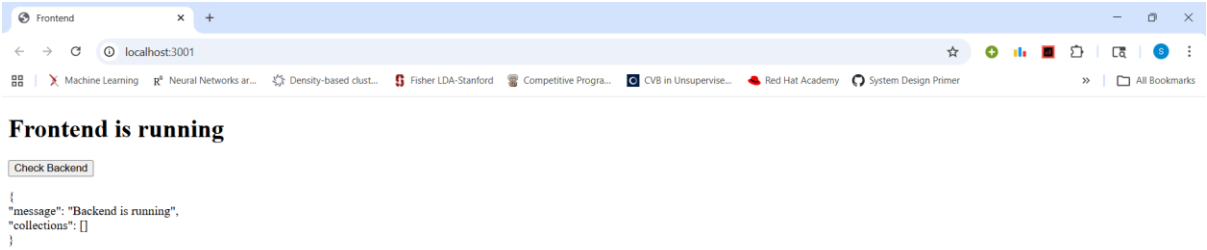
Frontend is running
Check Backend

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## Self-Note:

**Built a full-stack microservices application using Node.js, MongoDB, and Nginx, containerized with Docker Compose. Implemented a Nginx reverse proxy to route frontend requests to the backend, enabling seamless communication within the container network. Configured environment variables securely using .env and GitHub Secrets, and designed a CI/CD pipeline with GitHub Actions to automate testing, integration checks, Docker image building, and deployment. The setup ensures a scalable, production-like environment with end-to-end validation of services.**